G10-GATE-A-AUDIT-v1: Can C_PROVEN imply spatial structure required by backward/unique continuation?

ZFL-CX

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$$\boxed{G7 = FROZEN,\ G8 = CLOSED/BLOCKED,\ G9\ BLOCKED,\ G10\text{-}DESIGN = DONE\ (72612f7)}$$

$$\boxed{\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\},\ T^* < \infty}$$

$$\boxed{Q_A : \mathcal{C}_{PROVEN} \overset{?}{\Rightarrow} \text{ exact spatial structure for BU/UC}}$$

1 What theorems require - explicit checklist

We check two canonical families (no new hypotheses):

1. **ESS backward uniqueness** for $\partial_t + \Delta$: $|\partial_t u + \Delta u| \le C(|u| + |\nabla u|)$ in $Q = \mathbb{R}^3 \times (-1, 0]$, $|u(x,t)| \le \exp(C|x|^2)$, $u(x,0) = 0$ in $\mathbb{R}^3 \Rightarrow u \equiv 0$ in Q. For NS: applied to ω or u, needs differential inequality with bounded coefficients depending on u, and vanishing at final time.

2. **Unique continuation / Carleman**: If u satisfies Stokes-type inequality and $u = 0$ on open set or to infinite order at a point (x_0, t_0), then $u \equiv 0$. Requires vanishing of infinite order: $\forall k, |u(x,t)| \le C_k(|x - x_0| + |t - t_0|^{1/2})^k$ near (x_0, t_0).

In both cases required variable: u or ω or ∇u. Required region: all $\mathbb{R}^3$ at $t = 0$ for BU, or open set / point for UC. Required property: vanishing, smallness, decay, or bounded coefficients. Pressure must satisfy compatible condition. Need space-time window, not single t_n.

2 What C_PROVEN gives - point by point

1. Variable: C_F^{conc} gives $\|u(t_n)\|_{L^3(B_{r_n}(x_n))} \ge \varepsilon_0$. This is lower concentration, not upper smallness nor vanishing. Variable is u in L^3 local, not pointwise bound.

2. Region: Balls $B_{r_n}(x_n)$ shrinking $r_n \to 0$, $x_n \to x^*$. No information of $u = 0$ outside. No open set where $u \equiv 0$. No global decay. C_E^{weak} gives L^2 global but not pointwise $e^{C|x|^2}$ bound nor L^∞.

3. Vanishing vs lower bound: $\boxed{\text{lower concentration} \ne \text{vanishing order}}$. ESS BU requires $u(\cdot, 0) = 0$. We have opposite: $\|u(t_n)\|_{L^3(B_{r_n})} \ge \varepsilon_0$ and $t_n \uparrow T^*$. No $u = 0$ at blow-up time, no infinite-order vanishing. C_J^{neg} is loss of direction regularity, not gain of vanishing.

4. Smallness/decay: C_{L^3} gives $\|u(t)\|_{L^3} \le C$? Actually not bounded; only necessary that if $T^* < \infty$ then L^3 must blow up? Wait Leray: $\|u(t)\|_{L^3} \to \infty$? No, L^3 may stay bounded, C_{L^3} is critical norm condition? But we have at most global bound? Even if bounded, bound is upper, not

smallness. BU needs small coefficients or bounded, but vanishing hypothesis missing. C_I^{freq} gives $N_c \to \infty$, indicating high-frequency creation, opposite of smallness.

5. Pressure: $-\Delta p = \partial_i \partial_j (u_i u_j)$. With only $L^3(B_{r_n}) \geq \varepsilon_0$, no $L^{3/2}$ smallness or decay for p on large region. Carleman for NS requires pressure control.

6. Space-time window: Our info at discrete times t_n, balls B_{r_n}. BU/UC need inequality holding in parabolic cylinder $B_R \times (-S, 0]$ with uniform coefficients. Not available from $\mathcal{C}_{PROVEN}$ alone.

3 Gate A checklist result

- (1) variable: have u lower bound, need upper/vanishing: MISMATCH. - (2) region: have shrinking balls concentration, need global vanishing or open set zero: MISMATCH. - (3) vanishing: need $u = 0$ to infinite order: we have $\geq \varepsilon_0$: OPPOSITE. - (4) smallness/decay: need decay/smallness: have lower bound + frequency divergence: OPPOSITE. - (5) pressure: need control: have none from $\mathcal{C}_{PROVEN}$ sufficient. - (6) space-time: need cylinder uniform: have discrete t_n snapshots.

Thus: No spatial property of type required by BU/UC is derivable from $\mathcal{C}_{PROVEN}$.

$$\boxed{A = BLOCKED}$$

$$\boxed{\|u\|_{L^3(B_{r_n})} \geq \varepsilon_0 \not\Rightarrow u = 0 \text{ nor smallness nor vanishing nor } e^{C|x|^2} \text{ decay}}$$

$$\boxed{\text{lower concentration} \neq \text{upper structure for BU/UC}}$$

4 ZFL kill

First indispensable requirement for BU/UC is $u = 0$ in region or infinite-order vanishing or global exponential bound with final zero data. $\mathcal{C}_{PROVEN}$ gives none, gives opposite lower bound.

Therefore:

$$\boxed{G10\text{-}GATE\text{-}A = BLOCKED}, \quad \boxed{G10 = CLOSED/BLOCKED}$$

No need to open B-D. No G10-01.

$$\boxed{G7\ FROZEN \to G8\ CLOSED \to G9\ BLOCKED \to G10\ CLOSED/BLOCKED}$$

$$\boxed{STOP = PROGRESO - nostructure for continuation activatable from C_{PROVEN}}$$

If later new critical-scale bound or new derivable vanishing/smallness appears from $T^* < \infty$, G10 could be revisited with new $\mathcal{C}$, but with current $\mathcal{C}_{PROVEN}$ it is closed clean.